The Reference Landscape: Who Publishes the Benchmarks

> Part of the Functional Series. A benchmark is only useful if you know what it measures. This document lists the reference publications the platform is compared against, what each covers, and the scope conventions that decide whether two numbers can be compared at all. The comparisons themselves follow in the next three documents.

1. The references used

Every figure quoted in this section was read from the publication itself, in the edition named here. Where a secondary source disagreed with the publication, the publication was used.

Reference	Edition used	What it provides	Role in this series
IPCC Sixth Assessment Report (AR6): Working Group I (physical science), Working Group III (mitigation), Synthesis Report	WGI 2021, WGIII 2022, Synthesis 2023	Assessed warming to 2020, emissions in 2019, the TCRE range, remaining carbon budget from 2020, and emission-reduction pathways	The scientific baseline and the source of the pathway benchmarks
Global Carbon Budget 2025 (Global Carbon Project)	*Earth System Science Data* 18, 3211, 2026; first released November 2025	Fossil and land-use CO_2, atmospheric growth, sinks, a 2025 projection, remaining carbon budget	The reference for CO_2 emissions and concentration; the original source of the platform's OWID data
UNEP Emissions Gap Report 2025 ("Off target")	UNEP, 2025	Total greenhouse gases in 2024 by gas, the gap between pledges and pathways, and warming projections	The reference for total GHG emissions and for the "gap" to Paris-aligned pathways
Indicators of Global Climate Change 2025 (Forster et al.)	*Earth System Science Data* 18, 3889, published 11 June 2026	Human-induced warming, observed warming, remaining carbon budget, annual update of AR6 indicators	The most current assessment of human-induced warming and of the carbon budget
WMO consolidated global temperature	WMO news release on 2025, eight datasets	A consolidated annual global temperature from eight independent datasets	The agency-level temperature reference
Copernicus (ERA5), **Berkeley Earth**	*Global Climate Highlights 2025* (January 2026); *Global Temperature Report for 2025* (data as of early January 2026)	Individual temperature datasets	Single-dataset comparisons; Berkeley Earth is also the platform's own source

Reference	Edition used	What it provides	Role in this series
NOAA Global Monitoring Laboratory	Annual mean files, retrieved October 2026	Global-mean and Mauna Loa CO_2 concentration	The like-for-like concentration comparison
Climate Action Tracker	Global update, 13 November 2025	Warming projections for current policies and national pledges	An independent warming-outlook comparison

2. Six scope questions to ask of any number

Most apparent disagreements between sources are differences of definition. Before comparing two figures, check each of these:

Question	Why it matters	Example from this series
Which gases?	CO_2 only, or all greenhouse gases	OWID/GCB report CO_2; PRIMAP-hist and UNEP report CO_2e
Which warming-potential set?	AR5 and AR6 GWP-100 values differ slightly for CH_4 and N_2O	PRIMAP-hist uses AR5; UNEP and IPCC AR6 use AR6
Land use in or out?	Land-use CO_2 is large (about 4–5 $GtCO_2$ a year) and very uncertain	PRIMAP-hist excludes it; UNEP and IPCC include it
International aviation and shipping in or out?	About 1.2 $GtCO_2$ a year	PRIMAP-hist and the sum of countries exclude it; OWID's World row and UNEP include it
Netted or gross?	The Global Carbon Budget subtracts a cement carbonation sink of about 0.2 GtC (0.7 $GtCO_2$) from fossil emissions	See Emissions
Station or global?	Mauna Loa runs about 1.6–1.8 ppm above the global mean	See Temperature and Concentration

For temperature the equivalent questions are **which baseline** (1850–1900 or a recent period), **which dataset or combination of datasets**, **a single year or a multi-year average**, and **observed or human-induced** warming.

3. A note on uncertainty language

The IPCC uses calibrated terms. *Likely* means at least a 66% probability; *very likely* means at least 90%. A confidence interval from a regression, such as the platform's 95% interval on the emissions–temperature slope, is a different object. Sources also quote uncertainty at different confidence levels: UNEP and the IPCC use 90% ranges for emissions, WMO a 90% range for temperature, Berkeley Earth 95%. When two numbers are compared here, the interval each one carries is stated.

See also

- Emissions Compared with Benchmarks

- Temperature and Concentration Compared with Benchmarks
- Pathways, Carbon Budgets and Warming Outlooks
- Data Sources: how each platform series is built